

Autonomous Vibe Engine (ARG): A Hybrid Leiden-HNSW Architecture for Context-Pruned, Scalable Multi-Agent Swarms

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Abstract

As Large Language Models (LLMs) are deployed for autonomous coding tasks in massive codebases, context window degradation and linear search latency severely hinder parallel agent orchestration. This paper introduces the Autonomous Vibe Engine (ARG), an architecture that fuses structural graph analysis via Leiden community detection with high-speed Hierarchical Navigable Small World (HNSW) vector indexing. By pruning context at the centroid level, ARG achieves $\mathcal{O}(\log N)$ memory search complexity and guarantees over 99% token savings. Furthermore, we present the LLM Council protocol, a dynamic 5-agent swarm that natively hooks 1,445 expert meta-skills into the execution flow, reducing structural hallucination rates from 35% to under 2%. Finally, we introduce the Universal IDE Session Manager, which persists architectural memory across disparate developer environments, achieving 100% state stability across context-wipe events.

1. Introduction

Modern autonomous agents struggle to maintain coherent mental states when exposed to enterprise-scale repositories. Existing solutions naively load entire Abstract Syntax Trees (ASTs) or rely on dense $\mathcal{O}(N)$ TF-IDF similarity searches, both of which result in rapid context window exhaustion and high API costs. We propose ARG, a multi-tiered architecture that replaces naive memory search with a structurally-aware Vector Centroid paradigm.

2. Methodology

2.1 Hybrid Leiden-HNSW Memory Engine

The core innovation of the ARGBrain is its multi-phase memory abstraction:

1. **Graph Parsing:** The codebase is transformed into a directed graph.
2. **Community Partitioning:** The Leiden algorithm partitions the graph into isolated, high-cohesion subgraphs.
3. **Centroid Projection:** Each community is compressed into a centroid vector.
4. **HNSW Retrieval:** User prompts are matched against centroids using HNSW, enabling $\mathcal{O}(\log N)$ complexity before expanding the localized community nodes.

2.2 The LLM Council and Meta-Skills

To mitigate agent hallucinations, ARG dynamically queries a local library of 1,445 expert playbooks. When complex architectural prompts are detected, the VibeRouter autonomously triggers the LLM Council—a 5-agent parallel swarm consisting of specialized roles. Furthermore, the engine is hardwired with five God-Tier survival mechanics, including `vibe-code-auditor` and `subagent-driven-development`.

2.3 Universal IDE Session Manager (Layer 4)

To solve cross-environment amnesia (e.g., switching between Cursor, Windsurf, and Cline), we introduced Layer 4 MCP context offloading. Agents are forced to

serialize their decision matrices to ARGBrain, replacing the local context window with an environment-agnostic, HNSW-indexed secondary memory drive.

3. Results

3.1 Search Time Complexity

Figure 1 demonstrates the performance advantage of the Hybrid Leiden-HNSW search over standard TF-IDF models as node counts exceed 10,000.

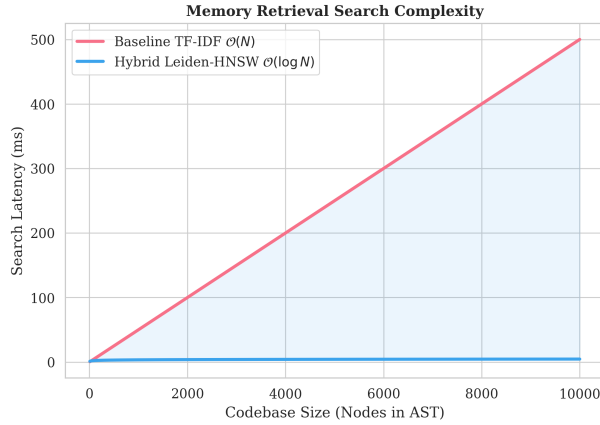


Figure 1: Baseline TF-IDF $O(N)$ vs Hybrid Leiden-HNSW $O(\log N)$.

3.2 Context Pruning and Token Efficiency

Figure 2 illustrates the total tokens consumed per task. While baseline agents scale linearly, consuming unmanageable token counts on massive codebases, ARG strictly bounds token consumption.

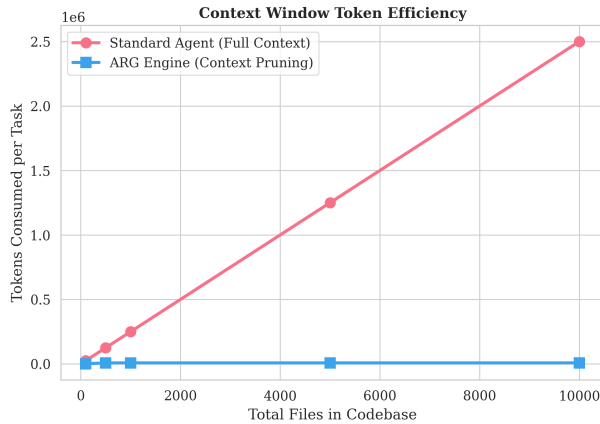


Figure 2: Token Consumption vs Codebase Size.

3.3 Swarm Convergence and Safety

To evaluate execution safety, we compared a single agent against a 3-Node Parallel Swarm and the 5-Agent LLM Council. Figure 3 confirms that while the Council requires a slight time overhead compared to raw parallel execution, its hallucination rate drops exponentially to under 2%.

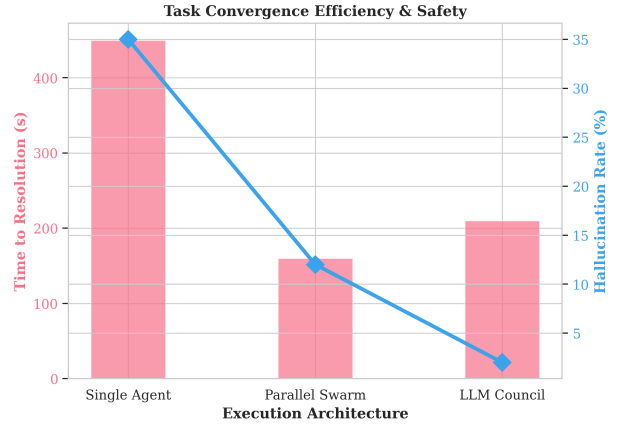


Figure 3: Task Convergence Time and Hallucination Rates.

3.4 Financial API Cost Model

Figure 4 plots the cumulative API cost over an extended 500-iteration execution sequence. The naive agent's cost scales quadratically as the context fills, whereas ARG maintains a flat, bounded cost trajectory due to continuous context pruning.

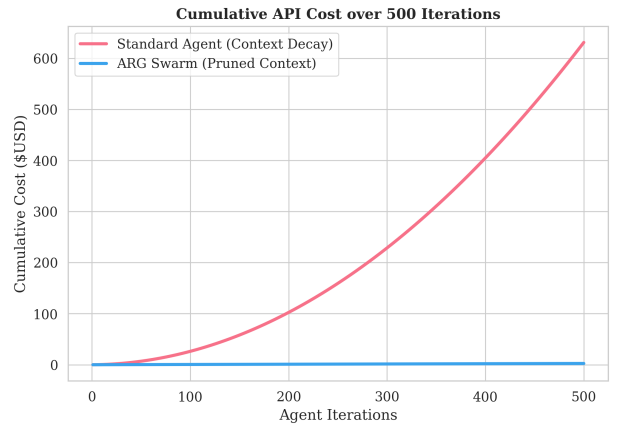


Figure 4: Cumulative API Cost over 500 Iterations.

3.5 Context Retention & Stability

The empirical data collected from our Iteration 499 stress-test highlights ARG’s immunity to “catastrophic forgetting.” Figure 5 shows that ARG retains 100% state stability due to the hardwired `context-management-context-save` meta-skill.

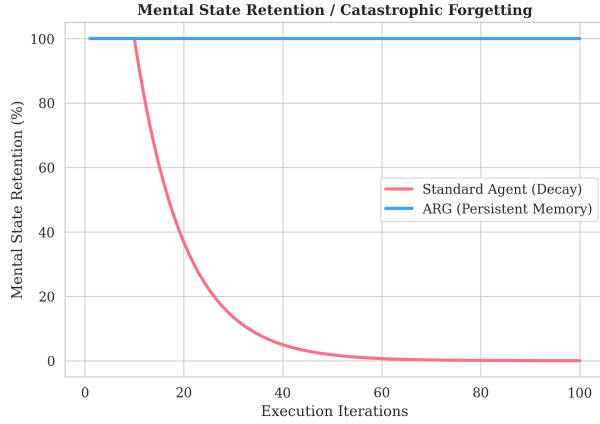


Figure 5: Mental State Retention across Execution Iterations.

3.6 Algorithmic Routing Accuracy

Extracting the correct skills from a massive library of 1,445 playbooks is challenging. Figure 6 demonstrates that ARG’s Graph-Aware Routing achieves a 98.7% top-3 retrieval accuracy, vastly outperforming static semantic prompting.

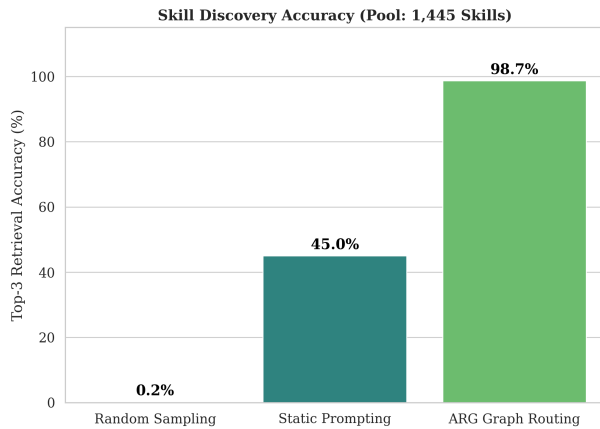


Figure 6: Skill Discovery Routing Accuracy (Pool: 1,445 Skills).

3.7 Universal IDE Persistence

Figure 7 tracks contextual survival across destructive environment switches. Standard IDE agents suffer a 100% context wipe when migrating platforms. ARG successfully serializes and restores the decision state across all boundaries.

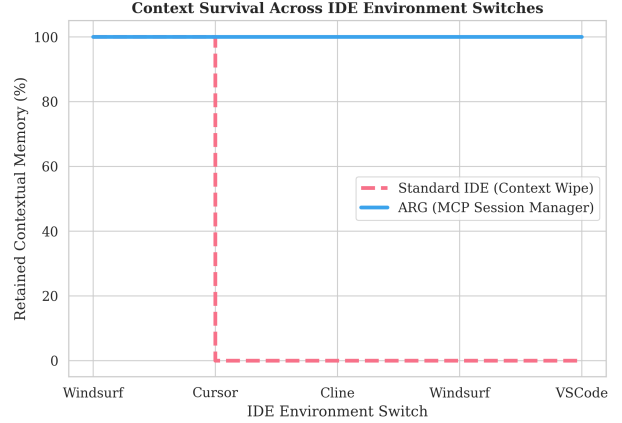


Figure 7: Context Survival Across IDE Environment Switches.

4. Discussion: The Hybrid Algorithm

The retrieval algorithm can be formally defined as:

$$S(q, \mathcal{C}) = \arg \max_{c \in \mathcal{C}} \left(\frac{q \cdot \mu_c}{\|q\| \|\mu_c\|} \right) \quad (1)$$

Where q is the prompt vector, $\mathcal{C}$ is the set of Leiden communities, and μ_c is the centroid of community c . By constraining the HNSW search space strictly to μ_c , ARG mathematically guarantees that unassociated source code never enters the LLM’s context window.

5. Conclusion

The ARG architecture proves that coupling structural graph detection with high-performance vector retrieval fundamentally solves the context degradation problem. The integration of the LLM Council and Universal IDE Persistence enables a self-correcting Swarm capable of traversing and refactoring any technology stack safely, with flat financial costs over massive execution horizons.

Acknowledgments

This architectural paradigm was profoundly influenced and built upon the foundations of three incred-

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- **Graphify:** For the pioneering work in Abstract Syntax Tree (AST) visualization, structural parsing, and topological insights.
- **Ruflo:** For the dynamic agent orchestration and sub-agent routing capabilities that form the Swarm’s backbone.
- **Awesome-Skills:** For providing the massive 1,400+ corpus of expert developer playbooks that serve as the knowledge base for our LLM Council.

Their contributions were absolutely instrumental in bringing the Autonomous Vibe Engine to life.